

Quadratic Equations (Algebra)

Rajat Kalia

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Infrastructure

#764, 41-A, Chandigarh {TC(official)} - 160036
 Shamlat (Ramgarh Sardaran), Ludhiana, Punjab
 SCO 24(TF), 41-D, Chandigarh - 160036
<https://www.alphaclasses.com/>

Educational

<https://github.com/rajatkalia1982>
<https://sourceforge.net/u/rajatkalia/profile>
<https://www.youtube.com/@rajatkaliaeducation>

Store

<https://www.lulu.com/spotlight/rajatkalia11>
<https://amazon.com/author/rajatkalia>

Sports

<https://www.youtube.com/@indiangamersorganization>
<https://www.facebook.com/groups/igohn>
<https://india.voobly.com/>

Social

<https://www.goodreads.com/sardarbhagatsingh>
<https://x.com/RajatKalia1>
<https://www.linkedin.com/in/rajatkalia007/>
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Blogs

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Freelancing

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<https://www.codecademy.com/profiles/rajatkalia>
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<https://www.scribd.com/user/308680256/Rajat-Kalia>

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Part I

Theory

Chapter 1

Quadratic Equations

1.0.0.1 Linked Comprehension type Problems

Comprehension 1: Let a, b, c be three distinct real numbers and $f(x)$ be a quadratic polynomial satisfying the equation

$$\begin{bmatrix} 9a^2 & 9a & 1 \\ 9b^2 & 9b & 1 \\ 9c^2 & 9c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(-2) \\ f(-3) \end{bmatrix} = \begin{bmatrix} 16a^2 + 10a \\ 16b^2 + 10b \\ 16c^2 + 10c \end{bmatrix}. \text{ Let } V \text{ be a point of local maxima of } y = f(x) \text{ and } A \text{ be the point where } y = f(x) \text{ meets the x-axis and } B \text{ be a point on } y = f(x) \text{ such that } AB \text{ subtends a right angle at } V$$

Q1: The curve $f(x)$ is given by

- A) $f(x) = -\frac{2}{9}x^2 + 2$
- B) $f(x) = -\frac{x^2}{10} + 16$
- C) $f(x) = -\frac{x^2}{16} + 10$
- D) None of these

Q2: The equation of chord AB can be

- A) $5x + 6y + 15 = 0$
- B) $6x + 5y + 15 = 0$
- C) $65x + 27y + 106 = 0$
- D) None of these

Q3: The area of the region lying between the curve and the chord AB is

- A) $\frac{3445}{8} \text{ unit}^2$
- B) $\frac{3445}{64} \text{ unit}^2$
- C) $\frac{3445}{128} \text{ unit}^2$
- D) None of these

Part II

Problem Set

Chapter 2

JEE Mains Problems

Q1: Let $\alpha, \alpha + 2, \alpha \in \mathbb{Z}$, be the roots of the quadratic equation $x(x + 2) + (x + 1)(x + 3) + (x + 2)(x + 4) + \dots + (x + n + 1)(x + n - 1) = 4n$ for some $n \in \mathbb{N}$. Then $n + \alpha$ is equal to :

[JEE Mains 2nd April 2026 Shift 1 Q1]

- A) 0
- B) 1
- C) 2
- D) 3

{Solution: <https://www.youtube.com/watch?v=AVSUt9a0>}

Q2 : Let α, β be the roots of the equation $x^2 - 3x + r = 0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 + 3x + r = 0$. If the roots of the equation $x^2 + 6x = m$ are $2\alpha + \beta + 2r$ and $\alpha - 2\beta - \frac{r}{2}$, then m is equal to :

[JEE Mains 2nd April 2026 Shift 2 Q1]

- A) -135
- B) -567
- C) 135
- D) 567

{ Solution : <https://www.youtube.com/watch?v=4sKXoUOd8Is> }